

WIRELESS EMERGENCY TRAFFIC SIGNAL PRIORITY SYSTEM

ARDUINO CONNECTIONS:

D2-D5 BUTTONS ARE FOR AMBULANCE

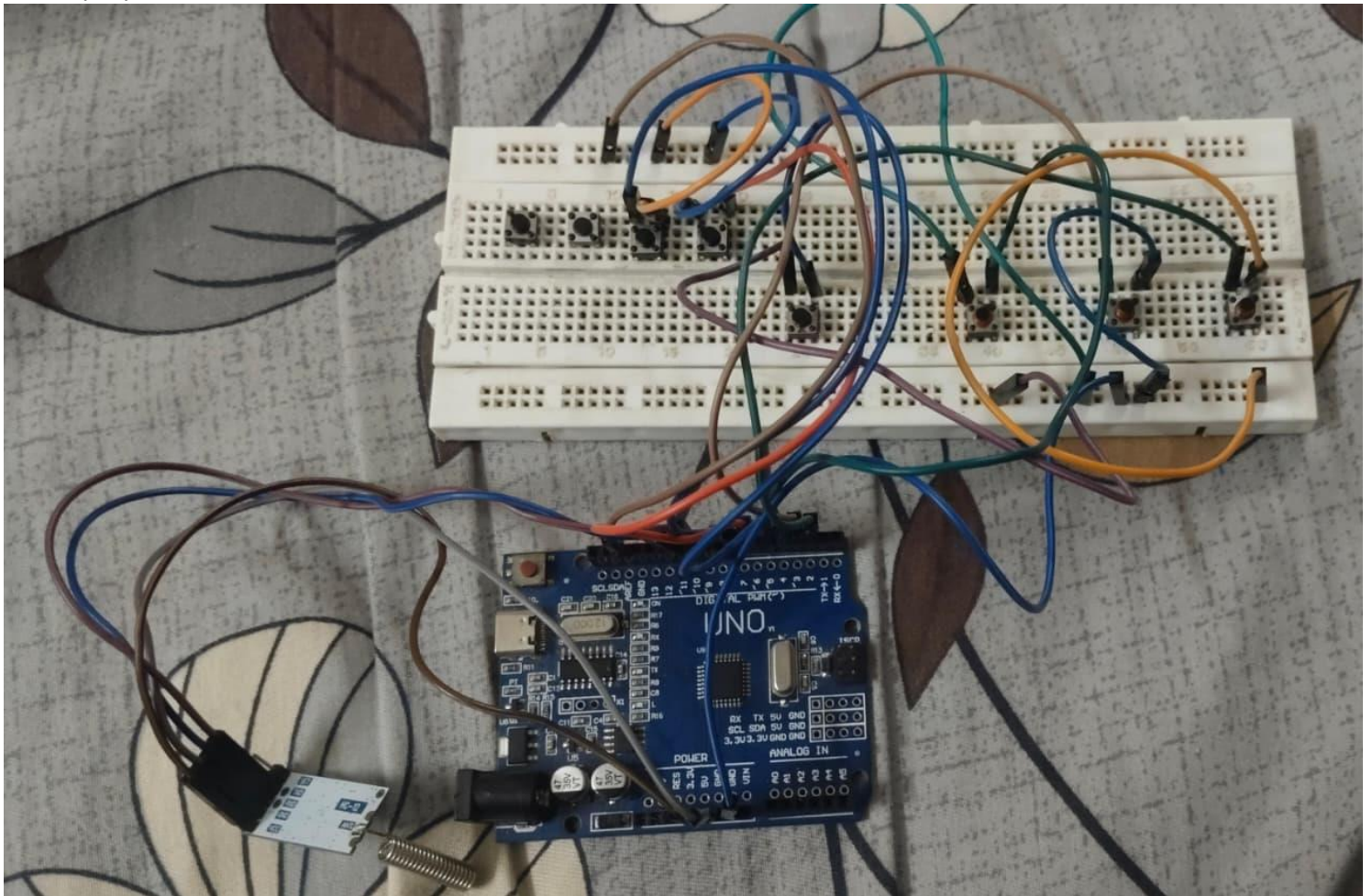
D6-D9 BUTTONS ARE FOR FIRETRUCK

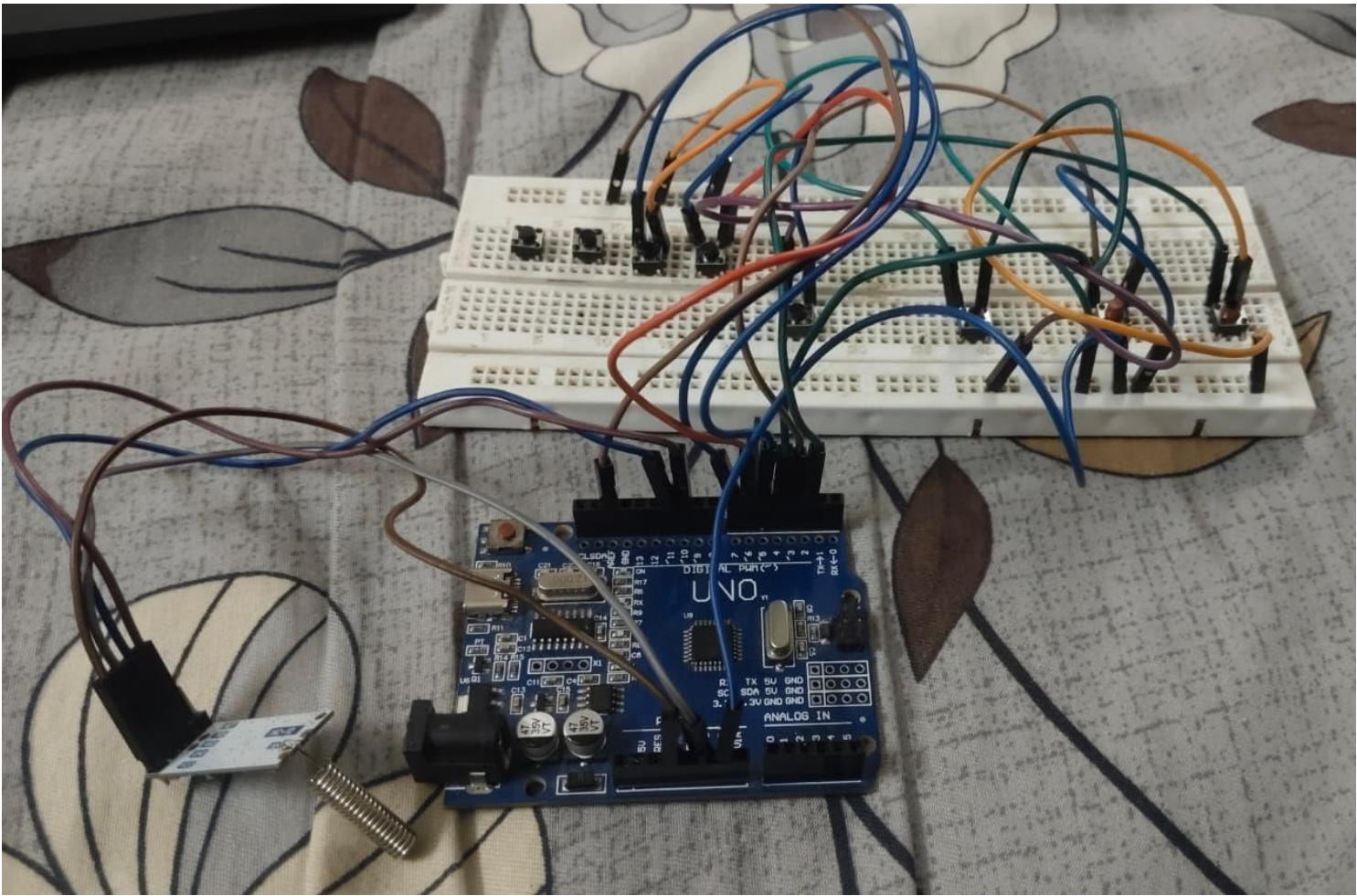
THESE PINS CONNECT TO ONE PIN OF THE BUTTON AND THE OTHER PIN SHOULD BE CONNECTED TO GND

HC12->VCC(3.3V) AND GND(0V)

HC12(RX)->D11

HC12(TX)->D10





STM32 CONNECTIONS:

DATA PIN CONNECTIONS:

PB0-PB11 CONNECT TO ONE END OF RESISTORS

OTHER END OF RESISTORS CONNECTED TO +VE TERMINAL OF LED(LONG TERMINAL)

-VE TERMINAL OF LED(SHORT TERMINAL) CONNECT TO GND

UART CONNECTIONS:

PA9(USART1 TX)->HC12(RX) AND PA10(USART1 RX)->HC12(TX) [BAUD RATE:9600]

PA2(USART2 TX) AND PA3(USART2 RX) CONNECTED INTERNALLY FOR PUTTY (DEBUGGING) [BAUD RATE:115200]
VCC(3.3V) AND GND(0V) CONNECTIONS NORMALLY.HC12 SUPPORTS ONLY 3.3V

STM32 IOC FILE CONFIGURATIONS:

***NVIC SETTINGS:** SWITCH ON ALL THE GLOBAL INTERRUPTS(TIM2 AND USARTS)*

***IN RCC:** HIGH SPEED CLOCK(HSE) SHOULD BE BYPASS CLOCK SOURCE FOR 72 MHZ CLOCK*

FOR TIM2 PARAMETER SETTINGS ARE: PRESCALER->71

COUNTER MODE->UP

COUNTER PERIOD->999

IN CLOCK CONFIGURATION CHANGE THE HCLK(MHZ) TO 72 THEN IT WILL AUTOMATICALLY CHANGES ALL THE VALUES.

THEN GENERATE CODE.

